INCH-POUND

MIL-C-71086 (AR)
29 April 1993
(See 6.6)

MILITARY SPECIFICATION

COMPONENT ASSEMBLY, A2, POWER SUPPLY & DRIVER: 12000021

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for Power Supply & Driver Component Assembly, A2, part of the Servo Amplifier used on Gun, Air Defense Artillery, Towed 20-MM, M167A1/A2. (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance, and Disposition of
MIL-W-63150 - Weapons and Support Materiel Standard Quality Assurance Provisions for

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12000021 - Component Assembly, A2, Power Supply & Driver
12950877 - Requirements for Soldering Electrical and Electronic Assemblies

SPECIAL PACKAGING INSTRUCTION

PS12000021 - Packaging of Component Assembly, A1, Power Supply & Driver: 12000021

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 12000021, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the Power Supply & Driver Assembly shall meet the requirements of MIL-W-63150 as follows:

a. Dimensions and tolerances
b. Inorganic protective surface finishings
c. Part identification and marking
d. Workmanship

3.3 Environmental service conditions.

3.3.1 Vibration. The Power Supply & Driver Component Assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to the equivalent of 6000 miles of track-laying vehicle operational vibration as defined by MIL-STD-810 narrowband random-on-random data for equipment installed on the deck of the M113 armored personnel carrier.

3.3.2 Shock. The Power Supply & Driver Component Assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ±10%, 11 milliseconds ±10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude Operating. The Power Supply & Driver Component Assembly shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg.).

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3.3.4 Temperature.

3.3.4.1 Operating. The Power Supply & Driver Component Assembly shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 degrees F and +155 degrees F.

3.3.4.2 Nonoperating (storage). The Power Supply & Driver Component Assembly shall not suffer permanent damage when subjected to the constant temperature extremes of -60 degrees F for a period of 24 hours and +160 degrees F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (nonoperating). The Power Supply & Driver Component Assembly shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the A2 portion of the schematic diagram for the Servo Amplifier.

3.5 Performance. All pin references apply to connector P1 of the Power Supply & Driver Component Assembly. Unless otherwise noted, all voltages are given with respect to ground (pin K).

3.5.1 Initial Setup. Attach 300 ohm loads to pins G and I. Attach 40 ohm loads to pins W and Y. Apply an input voltage of +28.0 ± 0.3 Vdc to pin E with the common at pin K.

3.5.2 DC output voltages. The DC output voltages shall be per Table I. The difference between the absolute values of the voltages at pins G and I shall be within 0.150 Vdc.

Table I. DC Output Voltages.

Pin	Voltage
O	+19.0 ± 1.5 Vdc
G	+15.0 ± 0.2 Vdc
I	-15.0 ± 0.2 Vdc

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3.5.3 Triangular wave output. The output at pin U shall be a triangular wave with a frequency of 625 ± 125 Hz, a positive peak at +2.50 to +2.80 Vdc, and a negative peak at +0.10 to +0.40 Vdc.

3.5.4 Input voltage dead band. The input voltage dead band at pin S shall range from plus and minus the value of the negative peak of the triangle wave observed in paragraph 3.5.3 with a ± 10% tolerance. When the input voltage at pin S is within this dead band, there shall be no modulated output at pins W and Y.

3.5.5 Modulated pulse width output. The modulated pulse width output shall be per Table II. The high level output voltage shall be 26.8 Vdc minimum.

TABLE II. Modulated Pulse Width Output

Input Voltage at Pin S	* Output Duty Cycle at Pin W	* Output Duty Cycle at Pin Y
+2.65 ± .03 Vdc	** 100% - 10%	0%
+2.05 ± .02 Vdc	** 75% ± 10%	0%
+1.45 ± .01 Vdc	50% ± 10%	0%
+0.85 ± .01 Vdc	25% ± 10%	0%
-0.85 ± .01 Vdc	0%	25% ± 10%
-1.45 ± .01 Vdc	0%	50% ± 10%
-2.05 ± .02 Vdc	0%	**75% ± 10%
-2.65 ± .03 Vdc	0%	**100% - 10%

* NOTE: Duty cycle = pulse duration/pulse repetition time.

** CAUTION: To prevent thermal damage do not operate at these duty cycles for more than 60 seconds.

3.5.6 Lockout time delay. At the 50% points, the lockout time delay between the application of +28.0 ± 0.3 Vdc to pin E and the corresponding fall in voltage at pins M and Q shall be 90 ± 30 milliseconds.

3.6 First article inspection. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the requirements of Note 2 of drawing 12950877.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-W-63150 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a. Temperature +50 to +90 degrees F

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b. Relative humidity 90% RH max.

c. Pressure 28 to 32 inches of Hg

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies listed below in the quantities indicated.

Name	Drawing	Quantity
PS & Driver Cmpnt Assy	12000021	3

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

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4.4.2.1 Components, assemblies and subassemblies.

a. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-W-63150 shall apply.

b. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.3). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

4.4.2.2 Final inspection. Subsequent to first article approval, examinations and tests related to Section 3 shall be performed as specified in 4.4.2.3 and 4.5.

4.4.2.3 Functional tests. The tests in Table III shall be inspected on a 100 percent basis.

TABLE III. Functional tests (100 percent inspection)

CHARACTERISTIC	REQUIREMENT	TEST PARAGRAPH
CRITICAL:	None	N/A
MAJOR:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
MINOR:	None	N/A

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4.5 Special sampling. One board assembly out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table III and shall then meet the requirements and tests in Table IV. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE IV. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PARAGRAPH
CRITICAL:	None	N/A
MAJOR:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
MINOR:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the

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component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-W-63150 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table V (Test equipment list) shall be utilized in the performance of the applicable test methods and procedures as specified in 4.7:

TABLE V. Test equipment list.

ITEM NUMBER	DESCRIPTION
1. Digital Voltmeter (DVM)	Minimum requirements: 5 1/2 digits, Vdc range: 0 to 300 V, Vdc accuracy: ± .02% + 5 counts at 300 mV range.
2. Oscilloscope	Minimum requirements: Analog/Digital Storage, dual trace, bandwidth: 20 MHz, sampling rate: 10MS/s, voltage range: 5mV/cm to 5 V/cm, sweep speed range: .1 usec/cm to .5 sec/cm analog mode and 20 usec/cm to .5 sec/cm digital storage mode.
3. Power Supply	Minimum requirements: Fixed, +28.0 ± 0.3 Vdc, 2.0 amp, load regulation: 1.0%.
4. Power Supply	Minimum requirements: Adjustable, -5.0 to +5.0 Vdc, 0 to 0.1 amp, load regulation: 0.1%.
5. Test Fixture	As specified in Figure 1. See Table VI for test fixture parts list.

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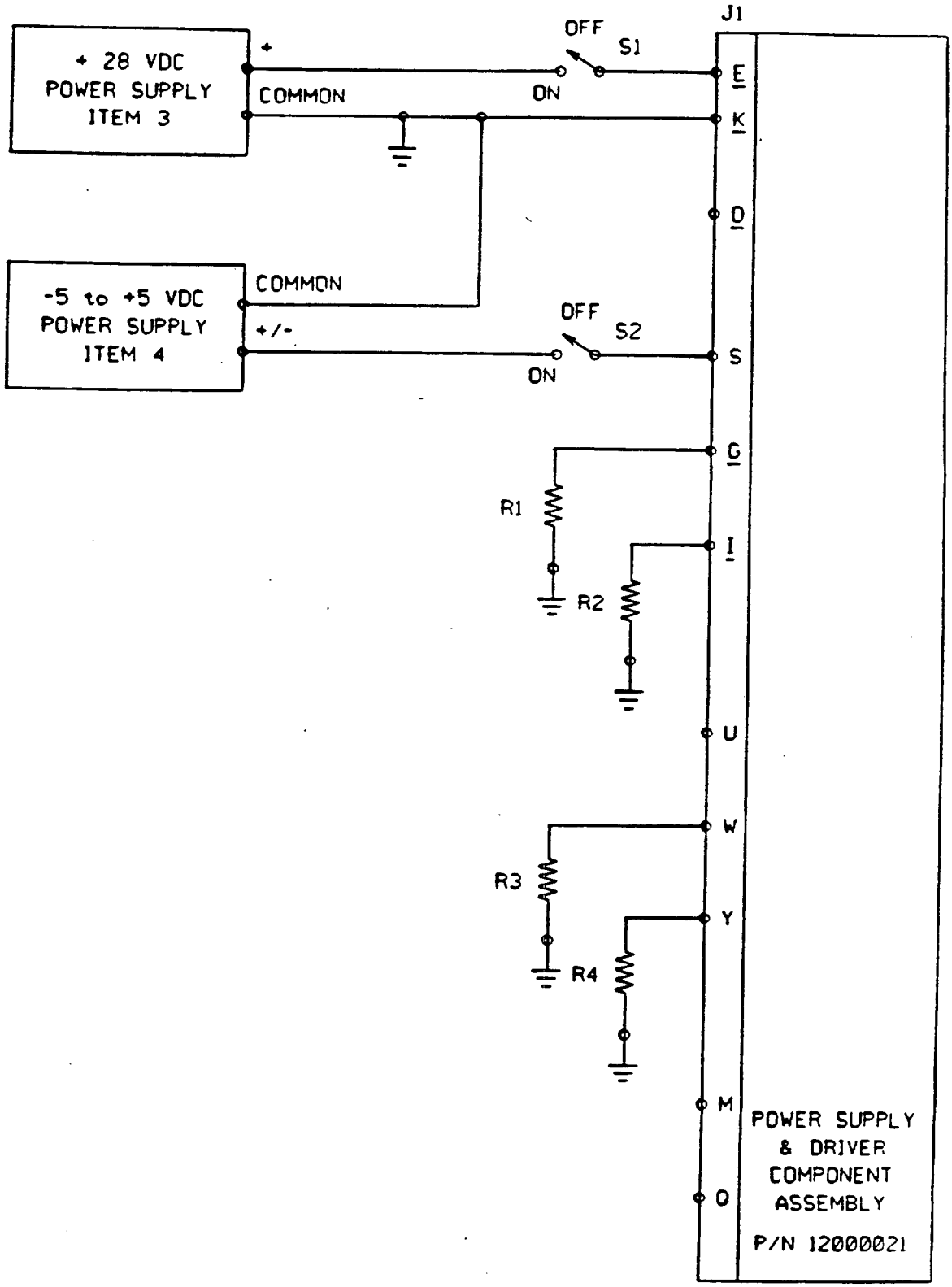


FIGURE 1. Power Supply & Driver Component Assembly, Test Harness

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TABLE VI. Test fixture parts list.

DESIGNATION	NOMENCLATURE	DESCRIPTION
R1 & R2	Resistors	300 ohm, 1%
R3 & R4	Resistors	40 ohm, 1%
S1 & S2	Switches	SPST
J1	Connector	M55302/6-05

Note: Resistors may be used in series to obtain the exact resistance value of the fixed resistors listed herein.

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in method 514, Category 8, Procedure I of MIL-STD-810 and 3.3.1 herein. In Step 1 of Procedure I only Steps 4 through 8 of paragraph II-2.1 apply. The power supply shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the power supply shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

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d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 Hertz.

After the application of all the shock impulses, the Power Supply & Driver Component Assembly shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating). This test shall be performed as specified in Procedures II, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and nonoperating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and nonoperating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (nonoperating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance. All pin references apply to connector P1 of the Power Supply & Driver Component Assembly. Unless otherwise noted, all voltages are given with respect to ground (pin K).

4.7.3.1 Initial Setup. Connect the Power Supply & Driver Component Assembly to the test fixture (item 5) with both switches set to the positions shown in Figure 1. Energize the power supplies (items 3 and 4) and set switch S1 to the ON position.

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4.7.3.2 DC output voltages. Verify that the Power Supply & Driver Component Assembly meets the requirements of 3.5.2.

a. Connect the DVM (item 1) between pin O and ground. The DC output voltage measured at pin O shall be +19.0 ± 1.5 Vdc.

b. Connect the DVM (item 1) between pin G and ground. The DC output voltage measured at pin G shall be +15.0 ± 0.2 Vdc. Record this voltage to the nearest millivolt.

c. Connect the DVM (item 1) between pin I and ground. The DC output voltage measured at pin I shall be -15.0 ± 0.2 Vdc. Record this voltage to the nearest millivolt.

d. The difference between the absolute values of the voltages recorded in 4.7.3.2.b and 4.7.3.2.c shall be within .150 Vdc.

4.7.3.3 Triangular wave output. Verify that the Power Supply & Driver Component Assembly meets the requirements of 3.5.3. Connect channel 1 of oscilloscope (item 2) to pin U and ground to pin K. The output measured at pin U shall be a triangular wave with a frequency of 625 ± 125 Hz, a positive peak at +2.50 to 2.80 Vdc and a negative peak at +0.10 to +0.40 Vdc. Record the voltage of the negative peak to the nearest millivolt.

4.7.3.4 Input voltage dead band. Verify that the Power Supply & Driver Component Assembly meets the requirements of 3.5.4.

a. Connect the DVM (item 1) between pin S and ground. Connect channel 1 of oscilloscope (item 2) to pin W, channel 2 to pin Y, and ground to pin K. Set switch S2 to the ON position.

b. Adjust the power supply (item 4) to provide an input voltage of +0.50 ± 0.01 Vdc at pin S and observe the modulated output (a repetitive pulse that occurs at the same frequency as the triangle wave observed in paragraph 4.7.3.3) at pin W.

c. Slowly decrease the voltage at pin S until the modulated output at pin W disappears. The voltage measured at pin S shall be within ± 10% of the negative peak voltage recorded in paragraph 4.7.3.3.

d. While monitoring the outputs at both pins W and Y, continue to slowly decrease the voltage at pin S until it reaches 0.00 ± 0.01 Vdc. No modulated output shall be observed.

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e. Adjust the power supply (item 4) to provide an input voltage of -0.50 ± 0.01 Vdc at pin S and observe the modulated output at pin Y.

f. Slowly increase the voltage at pin S until the modulated output at pin Y disappears. The absolute value of the voltage measured at pin S shall be within ± 10% of the negative peak voltage recorded in paragraph 4.7.3.3.

g. While monitoring the outputs at both pins W and Y, continue to slowly increase the voltage at pin S until it reaches 0,00 ± 0.01 Vdc. No modulated output shall be observed.

4.7.3.5 Modulated pulse width output. Verify that the Power Supply & Driver Component Assembly meets the requirements of 3.5.5.

a. Adjust the power supply (item 4) to provide an input voltage at pin S per the first value in Table VII.

b. There shall be no modulated output measured at pin Y (or W).

c. Measure the pulse duration and the pulse repetition time of the modulated output at pin W (or Y). The duty cycle of the modulated output at pin W (or Y) shall be per Table VII when calculated by using the formula:

Duty cycle = pulse duration / pulse repetition time

TABLE VII. Modulated Pulse Width Output

Input Voltage at Pin S	Output Duty Cycle at Pin W	Output Duty Cycle at Pin Y
+2.65 ± .03 Vdc	* 100% - 10%	0%
+2.05 ± .02 Vdc	* 75% ± 10%	0%
+1.45 ± .01 Vdc	50% ± 10%	0%
+0.85 ± .01 Vdc	25% ± 10%	0%
-0.85 ± .01 Vdc	0%	25% ± 10%
-1.45 ± .01 Vdc	0%	50% ± 10%
-2.05 ± .02 Vdc	0%	* 75% ± 10%
-2.65 ± .03 Vdc	0%	* 100% - 10%

* CAUTION: To prevent thermal damage do not operate at these duty cycles for more than 60 seconds.

d. The high level voltage of the modulated output measured at pin W (or Y) shall be 26.8 Vdc minimum.

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e. Adjust the power supply (item 4) to provide an input voltage at pin S per the next value in Table VII and repeat paragraphs b thru d above.

f. Set switches S1 and S2 to the OFF position.

4.7.3.6 Lockout time delay. Verify that the Power Supply & Driver Component Assembly meets the requirements of 3.5.5.

a. Connect channel 1 of oscilloscope (item 2) to pin E, channel 2 to pin M, and ground to pin K.

b. Set switch S1 to the ON position. The lockout time delay measured between the 50% point of the rise in voltage at pin E (when S1 is switched ON) and the 50% point of the corresponding fall in voltage that occurs at pin M shall be 90 ± 30 milliseconds.

c. Set switch S1 to the OFF position and connect channel 2 of the oscilloscope (item 2) to pin Q.

d. Set switch S1 to the ON position. The lockout time delay measured between the 50% point of the rise in voltage at pin E (when S1 is switched ON) and the 50% point of the corresponding fall in voltage that occurs at pin Q shall be 90 ± 30 milliseconds.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of Note 2 of drawing 12950877 as specified in paragraph 3.7.

PACKAGING

5.1 General. Preservation, packaging, and marking shall be in accordance with Special Packaging Instruction PS12000021. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Three servo amplifiers are used on each of the M163A1 and M167A1 weapon systems. One shall be used for elevation to drive the gun up or down. Two shall be used similarily for azimuth positioning of the gun.

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6.1.2 Functional Description. The Power Supply/Driver Component Assembly contains the following circuitry:

a. Pre-regulator to reduce the input voltage from +28 Vdc to +19.0 Vdc needed to supply the dc-dc converter and lockout delay cirduits.

b. Dc-dc converter to change the voltage from +19.0 Vdc single input to +17.1 Vdc dual output needed to supply the dual regulator.

c. Dual regulator to regulate the voltage from ±17.1 Vdc input to ±15.0 Vdc output needed for all signal level circuits.

d. Triangle wave oscillator ro produce a +2.4 Vp-p, 625 Hz triangle wave for input to the pulse width modulator.

e. Pulse width modulator to convert the signal from the lag-lead amplifier to cw or ccw pulses that have a duty cycle proportional to the voltage input.

f. Drive stages to increase the power level of the positive and negative output pulses from the pulse width modulator.

g. Delay circuits to provide the 90 msec delay of turret motion each time power is applied to the servo amplifier to allow electronic circuits in the system to stabilize.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.2).

c. Requirements for submission of first article sample.

d. Applicable stock number.

e. Packaging requirements, if other than specified in Section 5.

f. Serialization requirements, if applicable.

g. Certificate of conformance for each lot or shipment of product.

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6.3 Submission by the contractor.

a. Inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

b. Alternative quality conformance provisions for approval. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation to the technical activity responsible for the preparation of this specification.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Gun, Air Defense Artillery, Self-Propelled 20-MM, M163A1/A2 (VADS)
Gun, Air Defense Artillery, Towed 20-MM, M167A1/A2 (VADS)
Servo Amplifier
Vulcan Air Defense System (VADS)

6.6 Issue status. This issue is an initial public release for unlimited distribution and supersedes Section II - Power Supply/Driver Printed Circuit Board Assembly (A2) (12000021) of Performance Specification for Servo Amplifier (12000006), PE12000006 dated 16 June 1980.

Custodian:
Army-AR

Preparing activity:
Army-AR
(Project 1285-A871)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-71086	29 April 1993

3. DOCUMENT TITLE
COMPONENT ASSEMBLY, A2, POWER SUPPLY & DRIVER: 12000021

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
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8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
ATTN; SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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